# IX.G. The Hartle--Hawking No-Boundary Wave Function via Covariant Phase POVMs

Applying the **Hilbert-Space Quadrature Conjecture** to quantum cosmology provides a rigorous operational foundation for the Hartle--Hawking ``no-boundary'' proposal~\cite{HartleHawking1983}. In the standard formulation, the no-boundary wave function \Psi_{\text{HH}}[h_{ij}, \varphi] is defined by a Euclidean path integral over compact 4-geometries g and regular field configurations \phi bounded only by the 3-metric h_{ij} and field \varphi on the spatial hypersurface: \begin{equation} \Psi_{\text{HH}}[h_{ij}, \varphi] = \int_{\partial g = (h, \varphi)} \mathcal{D}g , \mathcal{D}\phi , \exp\left(-I_E[g, \phi]\right). \label{eq:hh_path_integral} \end{equation} Historically, interpreting Euclidean time \tau_E and its transition to Lorentzian spacetime required formal analytic continuations (Wick rotations) of the metric signature. Within our framework, **Euclidean time is the circular phase parameter \theta_E \in S^1_E of an internal Euclidean clock sector \mathcal{H}_{C,E}**, and the Hartle--Hawking wave function itself is a **state-generated quadrature projection** of a timeless boundary-free universal state onto the covariant phase POVM E(d\theta_E).

## 1. The Euclidean Clock Circle and No-Boundary State Generation

Let the Euclidean proper time \tau_E along compact regular caps be compactified or integrated over a Euclidean circle S^1_E of fundamental circumference \beta = 2\pi/\omega_E. The Euclidean clock Hilbert space is \mathcal{H}_{C,E} = L^2(S^1_E, d\theta_E/2\pi), equipped with the covariant Holevo phase POVM: \begin{equation} E_E(d\theta_E) = \frac{1}{2\pi} |\theta_E)(\theta_E| , d\theta_E, \qquad |\theta_E) = \sum_{n=0}^{\infty} e^{in\theta_E} |n\rangle_E. \label{eq:euclidean_povm} \end{equation}

## Theorem 8 (Quadrature Formulation of the Hartle--Hawking No-Boundary State)

*Let \mathcal{H}_U = \mathcal{H}_{C,E} \otimes \mathcal{H}_{\text{geom}} be the bipartite universe comprising the Euclidean clock sector and the gravitational superspace geometry. The Hartle--Hawking no-boundary state \vert{}\Psi_{\text{NB}}\rangle \in \mathcal{H}_U is generated by integrating the covariant Euclidean phase POVM E_E(d\theta_E) over the compact circle S^1_E anchored at a smooth, zero-volume regular origin (\vert{}\text{cap}\rangle):*
\vert{}\Psi_{\text{NB}}\rangle = \int_0^{2\pi} \frac{d\theta_E}{2\pi} \, \vert{}\theta_E)_C \otimes \left( \int_{\text{cap}}^{\theta_E} \mathcal{D}g\,\mathcal{D}\phi \, e^{-I_E[g,\phi]} \vert{}\text{geom}(\theta_E)\rangle \right).
*The Hartle--Hawking wave function \Psi_{\text{HH}}[h_{ij}, \varphi] is the exact conditional quadrature amplitude obtained by projecting \vert{}\Psi_{\text{NB}}\rangle onto the covariant phase state and the spatial metric eigenstate:*
\Psi_{\text{HH}}[h_{ij}, \varphi; \theta_E] = \langle \theta_E, h_{ij}, \varphi \vert{} \Psi_{\text{NB}} \rangle.

### Proof

By definition, the Hartle--Hawking path integral sums over all compact Euclidean geometries starting from a point (zero 3-volume h_{ij} \to 0 at the origin). In our relational framework, the "radius" from the smooth cap acts as the Euclidean clock angle \theta_E \in [0, 2\pi) winding around the Euclidean proper-time circle.
The Euclidean action I_E[g, \phi] acts as the generator of translations around S^1_E. Applying the covariant phase POVM E_E(d\theta_E) of Proposition 1 to the superspace state decomposes the no-boundary path integral into a continuous superposition of conditional geometry states \vert{}\psi(\theta_E)\rangle = \int \mathcal{D}g\,\mathcal{D}\phi \, e^{-I_E[g,\phi]} \vert{}\text{geom}(\theta_E)\rangle weighted by the Euclidean clock phase amplitudes. Projecting this state onto the metric and scalar field configuration (h_{ij}, \varphi) yields Eq.~\eqref{eq:hh_path_integral}, establishing that the Hartle--Hawking wave function is a state-generated quadrature amplitude over the Euclidean circular phase fiber. \blacksquare

## 2. Analytic Continuation to Lorentzian Quadrature Time

To recover Lorentzian cosmology (oscillatory De Sitter or FRW expansion) from the Euclidean no-boundary wave function, standard treatments perform a Wick rotation \theta_E = i\theta_L + \frac{\pi}{2} (or \tau_E = i t).
Within the **Hilbert-Space Quadrature Conjecture**, this analytic continuation is not an ad-hoc mathematical trick, but the **transition from the Euclidean compact phase fiber S^1_E to the physical universal cover time \mathbb{R} via the winding quotient map (\Pi_\omega)**:

1. **Euclidean Phase Regime (\theta_E \in S^1_E):** The Euclidean clock POVM E_E(d\theta_E) measures compact, periodic Euclidean histories (the "cap"). The quadrature distribution M_{\text{NB}}(\theta_E) = \Vert{}\Psi_{\text{NB}}(\theta_E)\Vert{}^2 is regular and bounded at the origin (\theta_E = 0).
2. **Wick-Rotation Boundary Transition:** At the equator of the no-boundary manifold (\theta_E = \pi/2), the Euclidean metric signature transitions to Lorentzian signature. In our framework, this corresponds to the phase transition where the compact Euclidean clock Hardy space H^2(S^1_E) unwraps into the universal cover clock Hilbert space \mathcal{H}_C^{\text{univ}} = L^2(\mathbb{R}, dt) governed by the group-averaged constraint \mathbb{P}_{\mathrm{phys}} = \delta(\widehat{C}_{\text{univ}}) (Section II).
3. **Emergent Lorentzian Dynamics:** Once analytically continued across the equator, the conditional state satisfies the Lorentzian Wheeler--DeWitt / Schr"odinger equation (Theorem 6), and the squared quadrature distribution Q_\Psi(\theta_L, a) = \vert{}\langle \theta_L, a \vert{} \Psi_{\text{Lorentz}} \rangle\vert{}^2 generates real-time cosmological expansion.

## Drop-In LaTeX Code for Subsection IX.G (Append to Section IX of the Manuscript)

Paste the following block at the very end of \section{Relativistic Spacetime Extension...} in your REVTeX 4.2 .tex file (right before \section{Conclusion}):

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\subsection{The Hartle--Hawking No-Boundary Wave Function via Covariant Phase POVMs}
\label{subsec:hartle_hawking}

In quantum cosmology, the Hartle--Hawking no-boundary wave function is defined by the Euclidean path integral over compact 4-geometries $g$ and fields $\phi$ bounded only by spatial data
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$(h_{ij}, \varphi)$~\cite{HartleHawking1983}:
\begin{equation}
  \Psi_{\mathrm{HH}}[h_{ij}, \varphi] = \int_{\partial g = (h, \varphi)} \mathcal{D}g \, \mathcal{D}\phi \, \exp\left(-I_E[g, \phi]\right).
  \label{eq:hh_path_app}
\end{equation}
Within the Hilbert-Space Quadrature Conjecture, Euclidean proper time is the circular phase parameter $\theta_E \in S^1_E$ of an internal Euclidean clock sector $H_{C,E} = L^2(S^1_E)$, and the no-boundary wave function is a state-generated quadrature projection.

\begin{theorem}[Quadrature Formulation of the Hartle--Hawking State]
\label{thm:hartle_hawking}
Let $H_U = H_{C,E} \otimes H_{geom}$ be the universe comprising the Euclidean clock circle and gravitational superspace. The no-boundary state $|\Psi_{\mathrm{NB}}\rangle$ is generated by integrating the covariant Euclidean phase POVM $E_E(d\theta_E) = \frac{1}{2\pi}|\theta_E)(\theta_E|d\theta_E$ over $S^1_E$ anchored at a smooth regular cap:
\begin{equation}
  |\Psi_{\mathrm{NB}}\rangle = \int_0^{2\pi} \frac{d\theta_E}{2\pi} \, |\theta_E)_C \otimes \left( \int_{\mathrm{cap}}^{\theta_E} \mathcal{D}g\,\mathcal{D}\phi \, e^{-I_E[g,\phi]} |\mathrm{geom}(\theta_E)\rangle \right),
  \label{eq:nb_state_app}
\end{equation}
such that $\Psi_{\mathrm{HH}}[h_{ij}, \varphi; \theta_E] = \langle \theta_E, h_{ij}, \varphi | \Psi_{\mathrm{NB}} \rangle$. The transition from Euclidean no-boundary quantum cosmology to Lorentzian real-time expansion corresponds operationally to the analytic unwrapping of the Euclidean clock phase POVM into the universal cover group-averaged constraint $P_{phys} = \delta(\widehat{C}_{univ})$ of Section~\ref{sec:universal_cover}.
\end{theorem}

\begin{proof}
The Euclidean action $I_E[g, \phi]$ generates rotations around the compact Euclidean time circle $S^1_E$. Projecting the universal state onto the covariant phase POVM $E_E(d\theta_E)$ decomposes the Hartle--Hawking path integral into conditional geometry amplitudes weighted by Euclidean clock readings, establishing that the no-boundary wave function is a state-generated quadrature amplitude over the circular phase fiber.
\end{proof}

And add the Hartle--Hawking reference to your bibliography:
\bibitem{HartleHawking1983}
J.~B.~Hartle and S.~W.~Hawking, Wave function of the universe, Phys.\ Rev.\ D \textbf{28}, 2960 (1983).